

Surface Treatment of the Lithium Boron Oxide Coated LiMn_2O_4 Cathode Material in Li-ion Battery

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Keywords: Li-ion battery, Cathode material, Lithium manganese oxide, Surface modification.

Abstract. Surface modification on the electrode has a vital impact on lithium-ion batteries, and it is essential to probe the mechanism of the modified film on the surface of the electrode. In this study, a $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ film was coated on the surface of the cathode material by solution method. The cathode powders derived from co-precipitation method were calcined with various weight percent of the surface modified glass to form fine powder of single spinel phase with different particle size, size distribution and morphology. The thermogravimetry/differential thermal analysis was used to evaluate the appropriate heat treatment temperature. The structure was confirmed by the X-ray diffractometer along with the composition measured by the electron probe microanalyzer. From the field emission scanning electron microscope image and Laser Scattering measurements, the average particle size was in the range of 7-8 μm . The electrochemical behavior of the cathode powder was examined by using two-electrode test cells consisted of a cathode, metallic lithium anode, and an electrolyte of 1M LiPF_6 . Cyclic charge/discharge testing of the coin cells, fabricated by both coated and un-coated cathode material, provided high discharge capacity. Furthermore, the coated cathode powder showed better cyclability than the un-coated one after the cyclic test. The introduction of the glass-coated cathode material revealed high discharge capacity and appreciably decreased the decay rate after cyclic test.

Introduction

In the past five years, with substantial growth in the electronic portable devices such as cellular phones, notebook, and personal digital assistance (PDA), compact lightweight batteries with high energy density and power capability have been focused. The Li-ion rechargeable batteries based on carbon as anode, non-aqueous electrolyte and cathode with the above-mentioned characterization have thus been extensively studied [1].

There are different types of cathode materials in Li-ion rechargeable batteries with low cost, high stability and good electrochemical performance, such as LiCoO_2 [2] and LiNiO_2 [3] with layer ($R\bar{3}m$) structure, LiNiVO_4 and LiCoVO_4 [4] with inverse spinel ($Fd\bar{3}m$) structure and LiMn_2O_4 [5] with spinel ($Fd\bar{3}m$) structure. Among cathode materials, LiMn_2O_4 is becoming the promising cathode material for commercial usage because of the lower cost than LiCoO_2 and more thermal stability than LiNiO_2 . However, the LiMn_2O_4 cathode material shows poor cycling behavior resulted from several factors. First, the Jahn-Teller distortion [6] caused by Mn^{3+} would lead to the degraded cycleability in LiMn_2O_4 during the Li^+ intercalation and de-intercalation. In our previous work, doping Li^+ [7] and transition metal like Cr^{3+} [8] could solve this problem successfully. However, as the cathode electrode is contacted with the non-aqueous electrolyte directly in Li-ion batteries, some side reactions would occur at the surface between the electrode and the electrolyte during the cycling test [9]. The surface modification of the cathode electrode is an efficient way to reduce the side reactions. The utilization of the metal oxides and inorganic materials for surface modification in both cathode and anode electrode [10-12] could improve the electrochemical property of Li-ion battery.

In this study, an effective approach was used to decrease the side reactions at the interface between

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the cathode electrode and the electrolyte and to satisfactory electrochemical property for Li-ion rechargeable batteries. The solution method was used to coat the LBO glass homogeneously on the surface of the cathode material. In addition, the influence of the LBO glass coating film on the electrochemical performance and structural stability of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material was examined.

Experimental Procedure

Powder preparation. The precursors of $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass, LiOH and H_3BO_3 , were mixed and stirred gently with the molar ratio of 1:2 in methanol. The precursor of LBO glass was then coated by solution method on the surface of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ powder which was synthesized in our previous work [7]. The solution was then heated at 50–60°C until the solvent was evaporated completely. The mixture powders of the precursor of the LBO glass and $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ were then calcined at 500°C for 10 hr. The weight percent of the LBO glass was between 0–1 wt% before heat-treatment, and was in the range of 0–0.5 wt% after calcined.

Characterization and analysis. The compositions of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material were analyzed with an inductively coupled plasma-atomic emission spectrometer (ICP-AES, Perkinelmer, Optima 3000 DV, U.S.A.), and were furthermore determined with an electron probe microanalyzer (EPMA, JXA-8800M, JEOL, Japan). Since the lithium could not be detected in the quantitative analysis of EPMA, the content of manganese and oxygen would be evaluated first through ZAF technique in EPMA and then the amount of lithium was obtained by difference and normalization approach. The crystal structure and phase of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material were determined with an X-ray diffractometer operated at 30KV and 20mA from 15° to 70° (Rigaku, D/MAX-B, Japan) with a wavelength of $\text{CuK}\alpha$ ($\lambda=1.5406\text{\AA}$). The particle morphology and particle size and distribution of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ powder were examined using a field scanning electron microscope (FESEM, JSM-6500F, JEOL, Japan) at an accelerating voltage of 15 kV and the laser scattering (Horiba, LA 300, Japan), respectively.

Electrochemical characterization. The electrochemical behavior of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode electrode was measured by using two-electrode test cells consisted of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ as cathode, metallic lithium anode and an electrolyte of 1M LiPF_6 in a 1:1 (volume ratio) mixture of EC/DMC. All procedures in fabricating the cells were carried out in a specially designed chamber with low oxygen pressure ($\text{O}_2 < 2$ ppm) and low moisture ($\text{H}_2\text{O} < 2$ ppm). Details of the set-up of the 2016 coin cell were reported elsewhere [8]. The constructed 2016 coin cells was then cycled within the potential range of 3.0 V to 4.2 V at 0.1 C rate for the first cycle and at 0.2 C rate from the second cycle.

Results and Discussion

Thermal analysis. In order to determine the chemical reaction of the precursor and the calcining temperature of the co-precipitation products, thermal analysis is necessary to provide the optimum heat-treated temperature.

The TGA trace of the decomposition of the precursor for the powder $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ at 2 °C min⁻¹ in air is presented in Fig.1. The TGA curve could be divided into three regions: (i) an initial weight loss of 5% around 130°C, (ii) a further weight loss of 25% between 200°C and 480°C, (iii) the final weight loss of 9% after 480°C. The region (i) of Fig.1 was assigned to the evaporation of free water absorbed from the atmosphere. The region (ii) could be contributed to the decomposition of the residual organics of LiOH and H_3BO_3 , which exhibit the melting point of 445 and 186°C, respectively, and then the precursor would react to form a LBO glass film on the surface of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ powder.

From the TGA trace, the total weight loss for the precursor of LBO is about 35 wt%. Moreover, the lowest calcined temperature of LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ to form the surface modified cathode material is supposed to be 480°C.

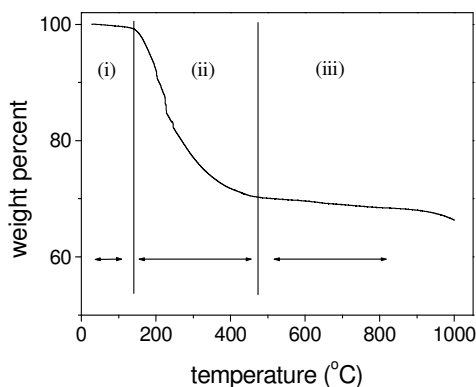


Fig. 1. TGA trace of the $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ precursor heat-treated from room temperature to 1000°C at rate of 2°C min^{-1} in air.

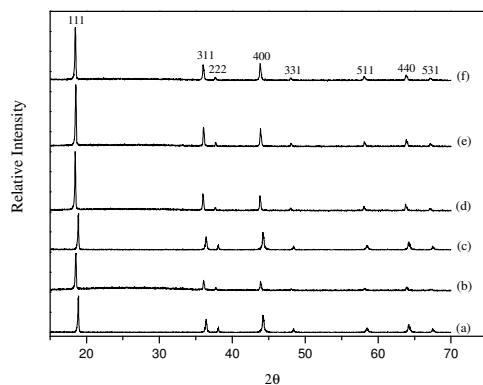


Fig. 2. XRD pattern of $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ coated with various wt% of LBO glass calcined at 500°C for 10 hr (a) uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$, (b) 0.1 wt%, (c) 0.2 wt%, (d) 0.3 wt%, (e) 0.4 wt%, and (f) 0.5 wt%.

Phase identification. The XRD patterns of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ coated with the various weight percent of LBO glass derived from solution method calcined at 500°C for 10 hr are shown in Fig. 2.

From Fig. 2(a), the XRD pattern of base $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ indicated that the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was the pure spinel cathode material without any impurity or other phases. In Figs. 2(b) to (f), it could be observed that the spectrum of the surface-modified $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was almost the same as the base one. From the absence of the peaks from other phases, it is suggested that the LBO glass was only coated at the surface of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and the diffusion of LBO into the matrix should be negligible.

On the basis of the XRD result, the lattice constant of LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was evaluated by a least-squares program, and plotted against the weight percent of the amount of LBO glass as indicated in Fig. 3. The lattice constant showed rather small deviation between $8.243 \text{ \AA}$ to $8.251 \text{ \AA}$, as compared with Amatucci's work [13]. This implies that the structure of the spinel cathode material was maintained after the process of surface modification, and the LBO was coated only on the surface rather than doped into the spinel structure.

Particle size and morphology. Figure 4 shows FE-SEM images of the uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$. In Fig. 4(a), the surface of the uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ with sharp edges was very smooth. However, in Fig. 4(b), the surface of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ with sharp edges was covered by a rough coating layer. The average particle size of the uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ measured by the Laser Scattering was in

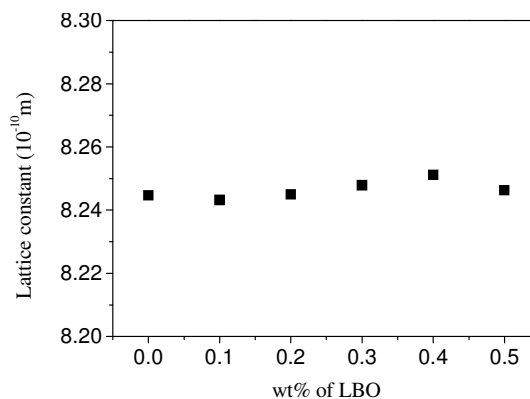


Fig. 3. Lattice constant vs. LBO wt% for LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$.

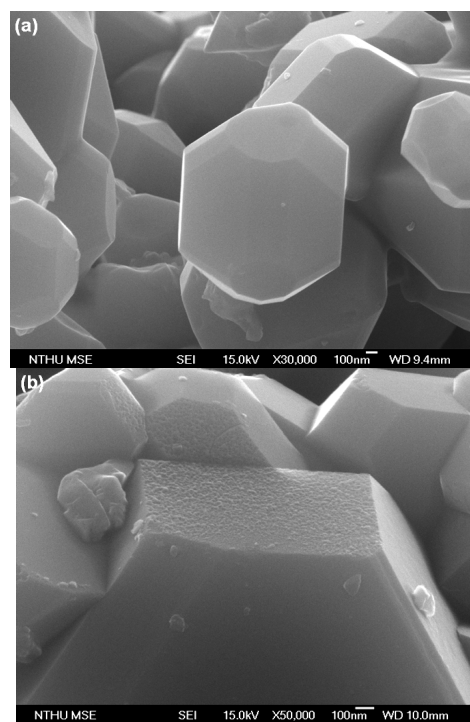


Fig. 4. FE-SEM images of (a) uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and (b) LBO-coated $\text{Li}_{1.08}\text{Mn}_2\text{O}_4$.

the range of 7-8 μm . The morphology of Fig. 4(a) to (b) is associated with that of single-crystal with a cubic structure. These particles with the amorphous LBO glass on the surface exhibit a well-developed octahedral structure, which is bounded by eight (111) planes [11].

Electrochemical property. The fading rate test of the base $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ calcined at 500°C for 10 hr, as shown in Fig. 5, was measured galvanostatically between 3 and 4.2 V with the assembly of $\text{Li}/\text{EC}+\text{DMC}+\text{LiPF}_6/\text{LiMn}_2\text{O}_4$ at 0.1 C rate for the first cycle and then at 0.2 C rate after the second cycle.

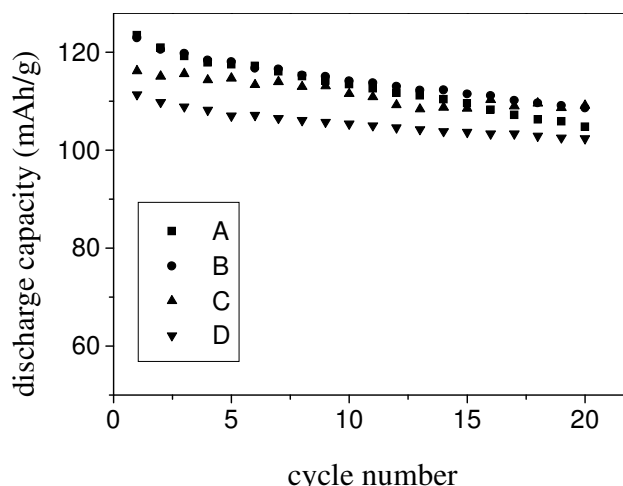


Fig. 5. Specific discharge capacity of various powders calcined at 500°C for 10 hr (a) uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$, (b) 0.1 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$, (c) 0.3 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and (d) 0.5 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$.

The uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ assembly delivered a value of 123.5 mAh/g at the first cycle and decayed to 104.8 mAh/g after 20 cycles, which showed a fading of 15.14%. The discharge capacity of the 0.1 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$, 0.3 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ and 0.5 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was 122.5, 116.2 and 111.4 mAh/g in the initial cycle, respectively, and remained 86%, 93% and 92% of its origin value after 20 cycles, respectively. With the increase of the amount in LBO glass, the fading rate of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material was obviously reduced. Among all the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ coated with various wt% of LBO glass, the 0.3 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material shows the best electrochemical performance. It is suggested that this is attributed to the decrease of the interface area between the electrode and the electrolyte. Hence, the side reaction leading to the electrolyte oxidation and dissolution of Mn_2O_3 is thus reduced [14].

The irreversible capacity of the first cycle of the uncoated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was 25 mAh/g calculated by the difference between 148.5 mAh/g of the first charge capacity and 123.5 mAh/g of the first discharge capacity. Furthermore, the 0.1 wt% LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ exhibits a value of 145 and 122.5 mAh/g for the charge and discharge capacity, respectively. An irreversible capacity of 22.5 mAh/g was lost during the first charge and discharge cycle. In addition, as compared with the first irreversible capacity of LiMn_2O_4 in our previous study [8], the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ possesses the lower charge capacity and higher discharge capacity.

The better electrochemical performance, like lower irreversible capacity, better cycling stability, lower charge voltage and higher discharge voltage of the surface-modified lithium manganese oxide cathode material may partially attribute to the $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass coating. In fact, the capacity loss of the conventional LiMn_2O_4 cathode material is proposed to the reaction of fluorinated anions with residual H_2O resulted in the formation of HF. The amount of HF contents is an important factor of the

MnO dissolution [13]. Hence, reducing the dissolution of Mn could be achieved by decreasing the HF contents at the surface of the cathode electrode. In literature, researchers [15, 16] pointed out that the $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass material was not proper in electronic conducting, but was a good conductor in transferring the Li^+ ion. Therefore, it is expected that the LBO layer with high Li^+ ionic conductor on the surface of the LiMn_2O_4 cathode material may encourage the Li^+ ion transferring through the interfaces between the LiMn_2O_4 cathode powders and lead to the decrease of the charge voltage and increasing the discharge voltage. The accomplishment in this research demonstrates that the LBO-coated LiMn_2O_4 cathode materials synthesized by solution method presents relatively good stability and electrochemical property.

Conclusions

1. $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass with various weight percent could be coated on the surface of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ cathode material successfully by solution method.
2. An amorphous $\text{Li}_2\text{O}-2\text{B}_2\text{O}_3$ glass film on the surface of the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ was observed by FESEM. The spinel XRD pattern of the LBO-coated $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ without the presence of other phases also indicated that the LBO glass was only coated on the surface rather than diffusing into the crystal.
3. The surface area between the cathode electrode and electrolyte could be reduced by surface-modification, and thus the electrolyte oxidation and the dissolution of Mn were depressed. Among all the $\text{Li}_{1+x}\text{Mn}_2\text{O}_4$ coated with various wt % of LBO glass, the 0.3 wt % LBO-coated one showed the best electrochemical performance.

Acknowledgement

The first two authors are grateful to the Coremax Taiwan Corporation, Taiwan for the financial support. Special thanks to General Manager Mr. Jim Ho for his kindness to endorse the project. Partial support from National Science Council, Taiwan, under the contract No. NSC-92-2216-E007-037 is also acknowledged.

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10.4028/www.scientific.net/KEM.280-283

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10.4028/www.scientific.net/KEM.280-283.671

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